

**COURSE TITLE : FLUID MECHANICS**  
**COURSE CODE : 3072**  
**COURSE CATEGORY : B**  
**PERIODS/ WEEK : 5**  
**PERIODS/ SEMESTER : 75**  
**CREDIT : 5**

**TIME SCHEDULE**

| MODULE       | TOPIC                               | PERIOD    |
|--------------|-------------------------------------|-----------|
| 1            | Properties of Fluids                | 20        |
| 2            | Fluid Friction and Flow Measurement | 15        |
| 3            | Transportation of Liquids           | 20        |
| 4            | Fluid Handling Equipments           | 20        |
| <b>TOTAL</b> |                                     | <b>75</b> |

**COURSE OUTCOMES :**

| SL.NO | SUB | STUDENT WILL BE ABLE TO                                                                           |
|-------|-----|---------------------------------------------------------------------------------------------------|
| 1     | 1   | To understand the properties of fluid.                                                            |
|       | 2   | To understand the laws of fluid statics and the working of pressure measuring instruments.        |
|       | 3   | To know the measurement of fluid flow and various fluid transportation machinery.                 |
|       | 4   | To understand the principles behind different flow meters.                                        |
| 2     | 1   | To install the flow meters and estimate the flow rate of fluid with flow meters                   |
|       | 2   | To understand different flow control devices and to gain the knowledge of using different valves. |
|       | 3   | To understand the principle working of equipments used for liquid transportation.                 |
| 3     | 1   | Able to calculate the energy requirement and efficiency of centrifugal pump.                      |
|       | 2   | Understand the working and application of blowers, compressors, vacuum pumps, jet ejectors.       |

**SPECIFIC OUTCOMES**

**MODULE– I PROPERTIES OF FLUIDS**

**1.1.0 Know the properties of Fluids**

- 1.1.1 Define density.
- 1.1.2 Compute specific gravity.
- 1.1.3 Define viscosity.
- 1.1.4 State Newton's law of viscosity.
- 1.1.5 Compute viscosity using Newton's law .
- 1.1.6 Classify fluids.

- 1.1.7 Compare Newtonian and Non-Newtonian Fluid.
- 1.1.8 Define viscosity index.
- 1.1.9 Determine viscosity using capillary tube method and Redwood viscometer.
- 1.1.10 solve simple problems using Newton's law.

**1.2.0 Understand the laws of fluid statics and the working of pressure measuring instruments.**

- 1.2.1 Define pressure.
- 1.2.2 Explain the laws of Fluid statics and its applications.
- 1.2.3 List the pressure measuring instruments.
- 1.2.4 Explain liquid column manometers.
- 1.2.5 Draw the sketch of differential manometer.
- 1.2.6 Describe 'U' tube manometer.
- 1.2.7 Draw the sketch of differential manometer.
- 1.2.8 Draw the sketch of an inclined tube manometer.
- 1.2.9 Derive an equation for measuring pressure difference for 'U' tube manometer.
- 1.2.10 compute pressure difference using manometers.
- 1.2.11 solve simple problems using above derived equations.
- 1.2.12 Define total pressure, gauge pressure, absolute pressure.
- 1.2.13 Compute total pressure from gauge pressure and atmospheric pressure.

**1.3.0 Apply the theories of fluid dynamics for computing energy of moving fluids**

- 1.3.1 List the types of flow.
- 1.3.2 Compare viscous and Turbulent flow.
- 1.3.3 Describe Reynold's experiment.
- 1.3.4 Define critical velocity.
- 1.3.5 Define Reynold's number.
- 1.3.6 State Transition range.
- 1.3.7 Solve simple problems using Reynold's number and predict types of flow.
- 1.3.8 Compute critical velocity.
- 1.3.9 Explain velocity distribution in pipes.
- 1.3.10 Explain the energies of fluid in motion.
- 1.3.11 Define fluid head.
- 1.3.12 State static head.
- 1.3.13 Describe impact head.
- 1.3.14 Define velocity head.
- 1.3.15 Define potential head.
- 1.3.16 Compute the total mechanical energy of a moving fluid.

**1.4.0 Apply Bernoulli's theorem**

- 1.4.1 State Bernoulli's theorem.
- 1.4.2 Explain Bernoulli's theorem.
- 1.4.3 State equation of continuity of liquid flow.
- 1.4.4 Explain rate of discharge.
- 1.4.5 Define mass rate.
- 1.4.6 Define volume rate.
- 1.4.7 Compute mass rate from velocity.
- 1.4.8 Explain the head developed by a pump in Bernoulli's equation.
- 1.4.9 Estimate the efficiency of a pump.
- 1.4.10 Estimate friction losses in a pumping system using Bernoulli's equation.
- 1.4.11 Apply Bernoulli's equation for finding the velocity at the discharge end.
- 1.4.15 Solve simple problems using Bernoulli's equation.

## **MODULE- II FLUID FRICTION AND FLOW MEASUREMENT**

### **2.1.0 Understand loss of head due to friction in fluids**

- 2.1.1 Explain skin friction.
- 2.1.2 Explain form friction.
- 2.1.3 Designate friction of straight pipe.
- 2.1.4 Determine friction losses in straight pipe in viscous flow using Hagen Poiseuille equation.
- 2.1.5 Estimate friction losses in straight pipes using Fanning's equation.
- 2.1.6 Define Fanning's friction factor.
- 2.1.7 Estimate friction factor in rough pipe.
- 2.1.8 Estimate friction factor in smooth pipe.
- 2.1.9 Explain the effects of Reynold's number and relative roughness on friction factor.
- 2.1.10 Compute friction losses in Bernoulli's equation.
- 2.1.11 Explain the friction losses in pipe fittings.
- 2.1.12 State expansion losses.
- 2.1.13 Compute losses due to expansion in pipes.
- 2.1.14 Compute losses due to contraction in pipes.
- 2.1.15 Solve simple problems.

### **2.2.0 Understand the working of flow measuring equipments.**

- 2.2.1 classify flow measuring equipments.
- 2.2.2 list the equipments in hydrodynamic method.
- 2.2.3 List the important mechanical flow measuring devices.
- 2.2.4 Define flow meter.
- 2.2.5 State the principle of flow meters.
- 2.2.6 Draw the sketch of an orifice meter.
- 2.2.7 Describe the working principle of an orifice meter.
- 2.2.8 Derive an equation for finding the velocity through orifice.
- 2.2.9 Draw the sketch of a venturi meter.
- 2.2.10 Describe the working principle of venturi meter.
- 2.2.11 Derive an equation for finding the velocity through the venturi meter.
- 2.2.12 Compare the advantages and disadvantages of orifice and venturimeter.
- 2.2.13 Estimate the volume rate through the pipe line using the equation for velocity through the orifice plate.
- 2.2.14 Estimate the volume rate through the pipe line using the equation for velocity through the venturi throat.
- 2.2.15 solve simple problems using above derived equations.

### **2.3.0 Comprehend Pitot Tube & Weir**

- 2.3.1 Draw the sketch of a pitot tube.
- 2.3.2 State the working principle of Pitot tube.
- 2.3.3 Plot the velocity profile using Pitot tube.
- 2.3.4 Derive an equation for finding the velocity.
- 2.3.5 Compute volume rate using equation for velocity through the pitot tube.
- 2.3.6 List the uses of weir.
- 2.3.7 Describe the rectangular Notch.
- 2.3.8 Describe V-Notch.

#### **2.4.0 Understand Rotameter as Displacement meter**

- 2.4.1 Explain the working principle of Rotameter.
- 2.4.2 Draw the sketch of Rotameter.
- 2.4.3 Describe the working of Rotameter.
- 2.4.4 List the types of floats in Rotameter.
- 2.4.5 Draw the sketch of a current meter.
- 2.4.6 Explain the working of a current meter.
- 2.4.7 Draw the sketch of a displacement meter.
- 2.4.8 Explain the working of a displacement meter.
- 2.4.9 Draw the figure of a wet gas meter.
- 2.4.10 Explain the working of a wet gas meter.
- 2.4.11 Describe an electromagnetic flow meter.

### **MODULE – III TRANSPORTATION OF LIQUIDS**

#### **3.1.0 Understand the Piping systems & Valves**

- 3.1.1 Differentiate between pipes and tubes.
- 3.1.2 List the advantages of transportation of fluid by pipe.
- 3.1.3 Define schedule number.
- 3.1.4 List the commonly used international standard for pipe and fittings.
- 3.1.5 Classify commonly used piping materials.
- 3.1.6 State the colour coding of pipes.
- 3.1.7 Explain the methods of joining pipes with figures.
- 3.1.8 List the advantages and disadvantages of screwed, flanged and welded joints.
- 3.1.9 Draw sketches of pipe fittings for screwed, flanged and welded pipe joints.
- 3.1.10 Explain the functions of gaskets and packing in pipe fittings.
- 3.1.11 List the gasket materials commonly used with their properties and uses.
- 3.1.12 Describe with sketch (1) stuffing box and (2) mechanical seal.

#### **3.2.0 Comprehend valves and valve materials**

- 3.2.1 Classify valves according to their functions.
- 3.2.2 List the common valve materials.
- 3.2.3 Describe with sketch Globe valve.
- 3.2.4 Draw the sketch of Plug cock valve.
- 3.2.5 Explain constructional details and working of Diaphragm valve.
- 3.2.6 Describe the working control valve.
- 3.2.7 Classify Gate valve.
- 3.2.8 Explain the constructional details and working of Gate valve with fig.
- 3.2.9 Describe the working of Ball valve with figure.
- 3.2.10 Explain the working of Butterfly valve.
- 3.2.11 Classify check valve.

### **MODULE – IV FLUID HANDLING EQUIPMENTS**

#### **4.1.0 Understand the pumping devices for liquids**

- 4.1.1 Explain the basis of classification of pumps.
- 4.1.2 Explain the functions of various types of Pumps in a Chemical Industry.

- 4.1.3 List the characteristic features of positive displacement and dynamic pumps.
- 4.1.4 Differentiate between piston pumps and plunger pumps.
- 4.1.5 With sketches explain the working principles of
  - (i) Piston pump – Simplex, Duplex and Single acting and double acting
  - (ii) Plunger pump – simplex pumps
- 4.1.6 List the field of application of piston and plunger pumps
- 4.1.7 Differentiate between rotary and reciprocating positive displacement pumps.
- 4.1.8 Explain the different types of reciprocating pump.
- 4.1.9 Explain the working of reciprocating pump.
- 4.1.10 List the advantages of reciprocating pump.
- 4.1.11 Describe with sketches of internal and external gear pump.
- 4.1.12 Explain, why safety relief valves are provided in piston pumps.
- 4.1.13 Explain the working of following pumps.
  - (i) Air lift (ii) Screw Pump (iii) Diaphragm pump
  - (iv) Acid egg (v) Solenoid pump (vi) Lobe Pump (vii) Vane pump
  - (viii) Electromagnetic pumps.
- 4.1.14 Explain the basic working principle of centrifugal pumps.
- 4.1.15 Classify centrifugal pump based on casing.
- 4.1.16 Explain different types of centrifugal pump impellers.
- 4.1.17 Explain the terms- Priming, Cavitations, NPSH
- 4.1.18 Compute the head of Centrifugal pump.
- 4.1.19 Explain the Performances of Centrifugal pump.
- 4.1.20 Explain the characteristic curves of centrifugal pump.

#### **4.2.0. Understand the Basics of Fans, Blowers and Compressors**

- 4.2.1 Explain the basics of classification of fans blowers and compressors.
- 4.2.2 Explain with sketches of radial flow and axial flow fans.
- 4.2.3 Describe with a sketch of a reciprocating compressor.
- 4.2.4 Explain the axial turbo compressor.
- 4.2.5 Explain the advantages of multi stage compressors.
- 4.2.6 Explain the effects of inter cooling in the efficiency of compressor.

#### **4.3.0 Understand the Vacuum Producing Equipments used in Industry**

- 4.3.1 Explain with a sketch of a Nash Hytor.
- 4.3.2 List the applications and advantages of Nash Hytor.
- 4.3.3 List the advantages of using steam jet ejectors for producing vacuum.
- 4.3.4 Explain with sketches of the following
  - (i) Single stage ejector.
  - (ii) Two stage ejector without condenser.
  - (iii) Multistage ejector with inter condensers.
- 4.3.5 Explain the use of condensers along with ejectors.

## **COURSE CONTENT**

### **MODULE– I PROPERTIES OF FLUID**

Properties of fluids – compressible - incompressible fluids – viscosity – definition-units - determination of viscosity. Viscosity of gases and liquids and their variation with temperature and pressure.-Viscosity index – Newtonian and Non-Newtonian fluids.

Laws of fluid statics and its application – fluid pressure – units and its measurements – concept of fluid heads and its calculations, gauge pressure – manometers – piezometer tubes, single column manometers, inclined manometers, simple U-tube manometer, two liquid manometer – construction, working and calculation of pressure difference based on reading of the above manometers, industrial type manometers.

Fluid dynamics: - Types of fluid flow in pipes, viscous and turbulent flow, Reynold's experiment, critical velocity, problems, types of fluid heads – static head, kinetic head, impact head, potential head, definitions and expressions.

Bernoulli's Theorem: - Application to simple problems, head developed by a pump – H.P of a pump and efficiency. Velocity distribution in pipes – relation between maximum and average velocity. Expressing flow rates as velocity, volume rate and mass rate. Equation for continuity.Simple problems.

### **MODULE– II FLUID FRICTION AND FLOW MEASUREMENT**

Causes of fluid friction in pipe systems – skin friction and form frictions – boundary layer separation and wake formation. (Description only)

Friction in straight pipe – Hagen Poiseuille equation for viscous flow (no derivation) and problems.

Friction in turbulent flow – Fanning's equation (derivation not required) friction factor chart- Effect of Reynold's number and relative roughness on friction factor- Empirical equation for friction factor- Calculation of friction losses in straight pipes and accounting for the loss in Bernoulli's equation. Friction losses in pipe fittings – equivalent length, expansion and contraction losses. Problems in fluid flow involving friction loss in pipe fittings. Economic pipe diameter.

Flow measurement: - Flow metering- classification of flow measuring equipments. Hydrodynamics method- mechanical method and special methods. Orifice meter- venturi meter - pitot tube and weirs. Basic working principle, derivation of mathematical expressions– simple problems.

Rotameter – working principles, materials of construction and applications. Mechanical flow measurements – Displacement meters, wet gas meter, rotating disc meter and piston meters, Current meters – cup current meters, propeller meters.

Special Methods: - Dilution methods, simple problems – Electro magnetic flow meter.

### **MODULE – III TRANSPORTATION AND CONTROL OF FLUIDS**

Pipes and pipe fittings: Pipes and tubes – pipe and tube standards,

Pipe fittings – fittings for screwed, welded and flanged joints.

Gaskets and packing: - Rubber, plastic, cork, rubber cork, asbestos, compressed asbestos fiber, metallic gaskets. Sealing of rotating shafts – stuffing boxes and mechanical seals .

Fundamentals of flow control mechanism and valve classifications-valve materials-Gate valve and its variations like sluice valve and slide valve- Plug valve (cocks) – 2 way and 3 way and non-lubricating plug valves- Constructional details of Ball valves – Globe valve- Butterfly valve Diaphragm valve.

## MODULE – IV FLUID HANDLING EQUIPMENTS

Pump classifications – Positive displacement and Centrifugal.

Positive displacement pumps – reciprocating and rotary – reciprocating type – piston plunger pumps – simplex and duplex – single acting and double acting.

Diaphragm pumps – Rotary – gear pump, screw pumps, lobe pumps.

Fluid displacement pumps – Airlift, Acid egg, Jet pumps, Electromagnetic pumps

Centrifugal pumps: - Basic working principles, types of impellers for different fluids, single suction and double suction type. Split case and multistage pumps. Priming of centrifugal pumps and self-priming pumps. Turbine pumps, NPSH, cavitations, selection of pumps based on liquid characteristic, head, capacity etc: -

Fans, Blowers, compressors – Positive displacement blowers, turbo blowers, turbo compressors, positive displacement compressors, vacuum pumps and ejectors. Nash Hytor. Installation, start up and shut down procedures of above machineries.

## REFERENCE

- |                                   |   |                               |
|-----------------------------------|---|-------------------------------|
| Walter.L.Badger&Julius.T.Banchero | - | Introduction to chemical Engg |
| Warren.L.McCabe&Julian.C.Smith    | - | Unit Operations               |
| K.A.Gavhane                       | - | Unit Operations – I           |